

equal.

- 7 (a) (i)** One volt (V) is the potential difference between two points in a circuit in which 1 joule of energy is converted when one coulomb passes from one point to the other.

**A common mistake made is that the answers merely refer to energy and charge without giving any units.*

- (ii)** e.m.f. of a source is the non-electrical energy converted into electrical energy, when unit charge is driven by the source round a complete circuit.

p.d. across a load is the electrical energy converted into non-electrical energy, when unit charge flows through the load.

(b) (i) $Q = It = (0.24)(5.0 \times 60) = 72 \text{ C}$

(ii) Energy transferred $= EQ = (1.5)(72) = 108 \text{ J}$

(iii) Energy loss in internal resistance $= I^2 rt = (0.24)^2 (0.25)(5.0 \times 60) = 4.32 \text{ J}$

Energy transferred in resistor R $= 108 - 4.32 = 104 \text{ J}$

- (iv)** Since the current through R is the same as that through the internal resistance, the ratio of their resistances is the same as that of their power and also that of their energy transferred.

$$R = \left(\frac{103.68}{4.32} \right) (0.25) = 6.0 \Omega$$

$$(c) (i) R_{total} = \frac{1.5 + 1.5}{0.41} = 7.317 \Omega$$

$$\text{New } R = R_{total} - 0.25 - 0.25 = 6.8 \Omega$$

(ii) A higher current causes more heating and an increase in the wire's lattice vibration, which results in more frequent collisions between the drifting electrons and the lattice ions. This increases the wire's resistivity.

(d) (i) The external loads will have a minimum total resistance of 3800Ω , while the total internal resistance is only 0.50Ω . This makes the resistance of the external loads at least 7600 times larger than the internal resistance, causing the internal resistance to be negligible.

**A statement that 'the internal resistance is small' is insufficient and is almost a paraphrase of the question.*

$$(ii) 1. V = \left(\frac{4000}{6000} \right) 3.0 = 2.0 \text{ V}$$

$$2. V = \left(\frac{1800}{3800} \right) 3.0 = 1.4 \text{ V}$$

(ii) In order for the p.d. across the fixed resistor to be 1.2 V at 0°C ,

$$\frac{R}{4000 + R} 3.0 = 1.2 \text{ V} \Rightarrow R = 2670 \Omega.$$

In order for the p.d. across the fixed resistor to be 2.4 V at 20°C ,

$$\frac{R}{1800 + R} 3.0 = 2.4 \text{ V} \Rightarrow R = 7200 \Omega.$$

There is no single value of R that satisfies both conditions.

**the resistances of the resistor were successfully calculated but many did not make any comment on their numerical answers.*